Lesson notes · for you

Networks and the defenders who watch them

Cyber security · Level 4 · Lesson 3

Time	Materials
18 min	A phone or computer for the lab; Large paper and pens for the network map

Learning goals

- Explain packets, addresses and routes.
- Name the three security goals and spot which one a problem breaks.
- Describe layers of defence for a real network.

Activity: Defend the network

1. Warm-up (3 min): Act out a photo's journey: three classmates are routers, and a fourth is the friend's phone. Pass four paper packets, then 'break' one router and find another route.
2. Lab (6 min): In the "Getting the message through" lab, break sensors one by one. Add relays until every message reaches the gateway again, and note the fewest relays you needed.
3. Map (6 min): In pairs, draw your campus's connection to the internet and mark where each layer of defence is or should be.
4. Share (3 min): Each pair suggests one improvement and names the security goal it protects.

Cut four 'packets' from paper, numbered 1 to 4. No real passwords or network names on the maps.

Discussion

- Why does the internet keep working when parts of it break?
- Which goal does a leaked course list break? A changed mark? A crashed website?
- Why is one strong defence not enough?

Check yourself

- One layer of defence, and the goal it protects.

Try it this week

In the lab, break a few sensors and add relays until every message gets through. Then draw your campus's or your home's network on paper, and mark where each layer of defence is or should be.

Worksheet

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Name _____ Class _____ Date _____

1 **On results day, the exam results website goes down for everyone. Which security goal is hit?**

◯ Integrity ◯ Availability ◯ Confidentiality

2 **Match each problem to the security goal it breaks.**

A course's student list with phone numbers is posted in a public group	• • • •	Availability
A fee record is quietly edited	• • • •	Confidentiality
The campus Wi-Fi is down during an online exam	• • • •	Integrity

3 **Put a photo's journey to your friend in order.**

☐ Each packet gets the sender's and receiver's addresses

☐ Your phone splits the photo into packets

☐ Your friend's phone puts the pieces back together

☐ Routers pass the packets along, hop by hop

4 **If one router in the middle of the internet breaks, your message can't be delivered.**

◯ True ◯ False

5 **Your university wants layers of defence for its network. Which of these are real layers?**

☐ Updates installed every month ☐ One long Wi-Fi password, written on the classroom wall

☐ Office computers on a separate network from students' Wi-Fi

☐ Turning off the logs to save space ☐ A firewall where the campus connects to the internet

✎ **Think of a service you use every day: a chat app, mobile money or the university's student portal. Which of the three goals would hurt you most if it failed, and why?**

_____

_____

_____

Check your answers

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1 **Availability**

Nothing was read by the wrong people (confidentiality) and nothing was changed (integrity).

2 **A course's student list with phone numbers is posted in a public group — Confidentiality**
A fee record is quietly edited — Integrity
The campus Wi-Fi is down during an online exam — Availability

Ask three questions: did the wrong people see it (confidentiality), was it changed (integrity), or could people not use it (availability)?

3 **Top to bottom: 2, 1, 4, 3**

Data is split and addressed before it leaves, routed hop by hop, and rebuilt at the end.

4 **False**

Networks are built with many routes, so packets go around a broken router.

5 **A firewall where the campus connects to the internet; Office computers on a separate network from students' Wi-Fi; Updates installed every month**

A firewall, separate networks and regular updates each stop a different kind of problem, so together they make layers.